

Alexandre Dalban

Citizenship: USA, EU | **Settled Status:** UK

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Education

CentraleSupélec, <i>MSc in Artificial Intelligence (M2)</i>	Paris, FR Sept 2025 – Nov 2026
Boston College, <i>BS in Mathematics and Computer Science</i>	Boston, USA Aug 2021 – May 2025

Experience

Applied AI Research Intern , <i>Siemens AG</i>	Munich, GER May 2026 – Nov 2026
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- Automating a manual step in manufacturing (CAD to CAM): built a model that automatically detects machinable features (holes, pockets, slots, etc.) on 3D part designs, replacing work currently done by hand. Uses self-supervised learning on sampled point clouds, building on Meta's Sonata and Xiaomi's Utonia.
- Experience using graph neural networks for classification.
- Current approach achieves ~95% panoptic quality, on par with state-of-the-art on the MFInstSeg benchmark.

Quantitative Finance Intern , <i>LeCap Asset Management (Hedge Fund)</i>	London, UK Summers 2024, 2025
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- Researched and validated 100+ alpha signals for systematic equity portfolio strategy, supporting portfolio construction.
- Built production Python tooling for the live trading environment, including an automated weekly reporting system evaluating 150+ signals across 8 performance metrics, replacing manual analysis.
- Built supervised forecasting pipeline spanning classical ML through transformer architectures (iTransformer), training on 90 proprietary indicators and Axioma factor-residualized returns across the top 1000 North American equities; best model achieved 54% daily directional accuracy vs. ~50% coin-flip baseline.
- Developed PPO reinforcement learning agents with reward based on rolling 5/10/20-day realized Sharpe and P&L, with custom cross-sectional and cross-temporal features (rolling autocorrelation, recent relative performance); agents output dynamic indicator weightings producing a composite prediction signal.

Projects & Academic Research

Domain-Specific LLM for Reliability Engineering [<i>GitHub</i>]	CentraleSupélec
<i>Supervised by Dr. Zhiguo Zeng (LGI lab); joint two-person project</i>	

- Built the first large training set for reliability engineering (failure analysis, MTTF, Markov chains, fault trees, etc.) by bootstrapping 56 textbook seeds into 866 verified Q&A pairs via a cross-model Self-Instruct pipeline, replacing the classical Self-Instruct same-model consistency filter, which we showed is trivially satisfied in technical domains.
- Led QLoRA fine-tuning of Qwen3-8B (Unsloth) and supported the co-author-led GRPO stage (DAPO-style dynamic sampling, verifiable rewards); ran 26 logged experiments on the HPC cluster (A100 GPUs, SLURM).
- Achieved +18.6 pp in-distribution accuracy (63.6% $\rightarrow$ 82.2%, $p = 0.002$, 10-fold cross-validation with Wilcoxon signed-rank test, every fold improved); paper in preparation for 2027 conference submission.

Phantom: Cryptocurrency Return Distributions Forecaster [<i>GitHub</i>]	Independent
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- Built a Transformer with a 2-layer cross-attention decoder jointly predicting Student- t return distributions across 1–30 day horizons for 362 cryptocurrencies using 120 day lookback; two-stage pipeline pretrained on 413 diverse assets (crypto, equities, FX, commodities), then fine-tuned on the crypto universe.
- Out-of-sample (Jan 2025–Mar 2026): rank IC = 0.124 at the 10-day horizon, robust to overlap-adjusted inference, with monotonic quintile sorts and positive IC in 15/15 months.
- Ran 8-version ablation isolating each design choice: synthetic SDE pretraining transferred zero real-world signal, multi-asset pretraining alone was directionally flat, and 4-hour bars degraded results due to sample overlap; self-audited the evaluation, correcting a $\sqrt{10}$ Sharpe-inflation error and publishing survivorship caveats; paper draft in ACM format.

Technical Skills

Languages: **Programming:** Python (advanced) | **Human:** Native French and English

Deep Learning: PyTorch, TensorFlow, Hugging Face, Sentence-Transformers, Cross-Encoders

Architectures: Transformers/attention, VAEs, GNNs, self-supervised & generative models, time-series forecasting

Quantitative Finance: Alpha research, factor models (Axioma residualization), systematic trading, signal evaluation (rank IC, Newey-West, Deflated Sharpe/PBO), long-short strategies, return distribution forecasting (Student- t), options pricing (Black-76, IV surfaces)

Data Science: Pandas, NumPy, Scikit-learn, LightGBM, Optuna, Apache Spark, SciPy, FAISS, Matplotlib

LLM Training & RL: QLoRA, GRPO, DPO, PPO, MaskablePPO, MCTS, Unsloth, synthetic data generation (Self-Instruct)

Tools: Claude Code (agentic coding), Git, asyncio/aiohttp, WebSocket/REST APIs, Docker, SQLite, MILP (PuLP), SLURM, LaTeX

Achievements: 1st in CentraleSupélec GreenAI hackathon, 2nd in HackEurope Solana prize (\$1000, Feb 2026), 13th in France at IMC Prosperity 4 (458th of 18,803 teams globally)